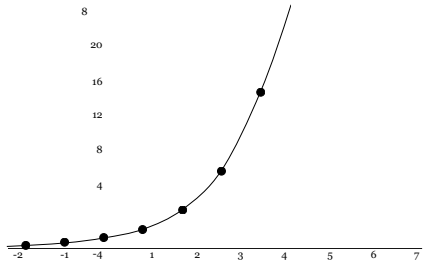
1.22 Exponentials and Logarithms

Chapter 14 · Original slide 4 · y={2}^{x}



x	-2	-1	0	1	2	3	4
y							

Note: Ensure that you can distinguish between a x^a (e.g. polynomial) term and an a^x exponential term. In the former the variable is in the base, and in the letter the variable is in the power. 2^x behaves very differently to x^2, both in its rate of growth (i.e. exponential terms grow much faster!) and how it differentiates.

Why are exponential functions important?

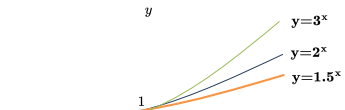
Each of the common graphs have a key property that makes them useful for modelling.

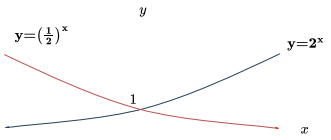
For reciprocal graphs $y = \frac{k}{x}$, as x doubles, y halves. This means we'd use it to represent variables which are inversely proportional.

Linear graphs are used when we're adding the same amount each time.

In contrast, exponential graphs are used when we're multiplying by the same amount each time. For example, we might use $1000(1.05^t)$ to model our savings with interest, where each year we have 1.05 times as much, i.e. with 5% added interest.

Chapter 14 · Original slide 5 · Contrasting exponential graphs





On the same axes sketch $y = 3^x, y = 2^x, y = 1.5^x$

On the same axes sketch $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$

Three important notes:

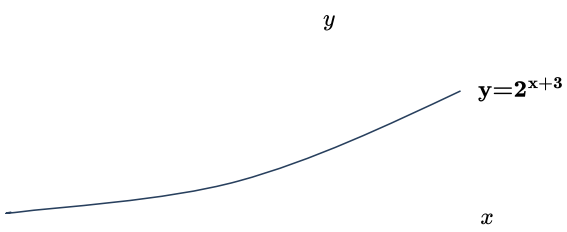
• $y = 2^x$ is said to be "exponential growing" whereas $y = \left(\frac{1}{2}\right)^x$ is said to be "exponentially decaying", because it's getting smaller (halving) each time x increases by 1.

• $y = \left(\frac{1}{2}\right)^x$ is a reflection of $y = 2^x$ in the line $x = 0$.
Proof:
If $f(x) = 2^x$, $f(-x) = 2^{-x} = \frac{1}{2^x} = \left(\frac{1}{2}\right)^x$

• $\left(\frac{1}{2}\right)^x$ would usually be written 2^{-x}.
You should therefore in general be able to recognise and sketch the graph $\mathbf{y=a^{-x}}$.

1.22 Exponentials and Logarithms

Chapter 14 · Original slide 6 · Graph Transformations



Sketch $y = 2^{x+3}$

The 'change' to 2^x is 'inside the function' (i.e. input x is replaced with $x + 3$).

Ensure you work out the new y-intercept.

Chapter 14 · Original slide 7 · Exercise 14A (Pages 313-314)

6 The graph of $y = ka^x$ passes through the points $(1, 6)$ and $(4, 48)$. Find the values of the constants k and a.

7 The graph of $y = pq^x$ passes through the points $(-3, 150)$ and $(2, 0.048)$.

a By drawing a sketch or otherwise, explain why $0 < q < 1$.

b Find the values of the constants p and q.

1.22 Exponentials and Logarithms

Chapter 14 · Original slide 21 · Logarithms

You know the inverse of many mathematical operations; we can undo an addition by 2 for example by subtracting 2. But is there an inverse function for an exponential function?

Function

12

Inverse

4

$x \times 3$

4

4

7

4

$x + 3$

4

x^2

16

4

4

x^5

1024

4

4

3^x

81

4

Such functions are known as logarithms, and exist in order to provide an inverse to exponential functions.

Chapter 14 · Original slide 22 · Interchanging between exponential and log form

$$3^2 = 9 \iff \log_3 9 = 2$$

! $\log_a n$ (“said log base a of n”) is equivalent to $a^x = n$. The log function outputs the missing power.

$$\log_2 8 = 3$$

Here are two methods of interchanging between these forms.

Pick your favourite!

Method 2: Do same operation to each side of equation.

Method 1: ‘Missing Power’

• Note first the base of the log must match the base of the exponential.

• $\log_2 8$ for example asks the question “2 to what power gives 8?”

Since KS3 you’re used to the idea of doing the same thing to each side of the equation that ‘undoes’ whatever you want to get rid of.

$3x + 2 = 11$

$3x = 9$

1.22 Exponentials and Logarithms

Chapter 14 · Original slide 23 · Examples

$\log_5 25 =$

Think: "5 to the power of what gives you 25?"

$\log_3(\frac{1}{27}) =$

$\log_3 81 =$

$\log_2(\frac{1}{16}) =$

$\log_2 32 =$

$\log_{10} 1000 =$

$\log_a(a^3) =$

$\log_4 1 =$

! $\log_a 1 = 0$ for all a.

$\log_4 4 =$

$\log_4(-1) =$

$\log_2(\frac{1}{2}) =$

Chapter 14 · Original slide 24 · With Your Calculator

There are three buttons on your calculator for computing logs:

$\log_\square \square$

$\log_3 7 =$

$\ln 10 =$

$\log_5 0.3 =$

$\ln e =$

ln

ln is the "natural log of **x**", meaning "log to the base e", i.e. it the inverse of e^x.

$\ln(x) = \log_e(x)$

We will use it more extensively later this chapter.

log

$\log 100 =$

Just like the $\sqrt{}$ symbol without a number is $\sqrt[2]{\square}$ by default, log without a base is base 10 by default when used on your calculator (although confusingly "log" can mean "ln" in mathematical papers)

1.22 Exponentials and Logarithms

Chapter 14 · Original slide 25 · Exercise 14D – Y1 Page 321

6 a Without using a calculator, justify why the value of $\log_2 50$ must be between 5 and 6.

b Use a calculator to find the exact value of $\log_2 50$ to 4 significant figures.

8 a Find the values of:

i $\log_2 1$ **ii** $\log_3 1$ **iii** $\log_{17} 1$ **b** Explain why $\log_a 1$ has the same value for all positive values of a $(a \neq 1)$.

7 a Find the values of:

i $\log_2 2$ **ii** $\log_3 3$ **iii** $\log_{17} 17$ **b** Explain why $\log_a a$ has the same value for all positive values of a $(a \neq 1)$.

Chapter 14 · Original slide 26 · Extension Exercise

1

[MAT 2015 1J] Which is the largest of the following numbers?

A) $\frac{\sqrt{7}}{2}$ B) $\frac{5}{4}$ C) $\frac{\sqrt{10!}}{3(6!)}$

D) $\frac{\log_2 30}{\log_3 85}$ E) $\frac{1+\sqrt{6}}{3}$

Non-calculator!

2

[MAT 2013 1F] Three positive numbers a, b, c satisfy

$\log_b a = 2$

$\log_b(c-3) = 3$

$\log_a(c+5) = 2$

This information:

specifies a uniquely;

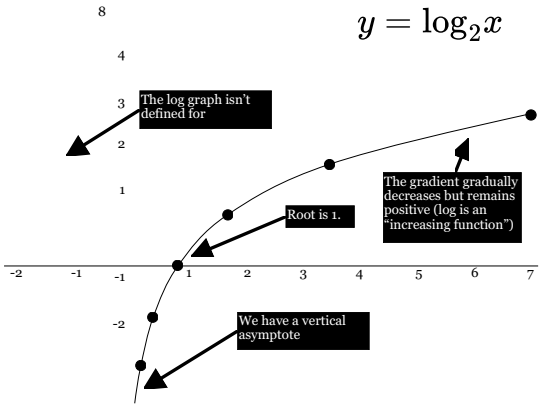
is satisfied by two values of a;

is satisfied by infinitely many values of a;

is contradictory

1.22 Exponentials and Logarithms

Chapter 14 · Original slide 27 · Exponentials and logarithms



x	0.25	0.5	1	2	4	8
y						

Chapter 14 · Original slide 28 · Laws of Logs

! Three main laws:

$\log_a x + \log_a y = \log_a xy$
$\log_a x - \log_a y = \log_a \left(\frac{x}{y}\right)$
$\log_a \left(x^k\right) = k\log_a x$

Special cases:

$\log_a a = 1 (a > 0, a \neq 1)$
$\log_a 1 = 0 \, (a > 0, a \neq 1)$
$\log \left(\frac{1}{x}\right) = \log \left(x^{-1}\right) = -\log (x)$

Not in syllabus (but in MAT/PAT):

$\log_a b = \frac{\log_c b}{\log_c a}$

The logs must have a consistent base.

i.e. You can move the power to the front.

We often try to avoid leaving fractions inside logs.

So if the answer was:

$\log_2 \left(\frac{1}{3}\right)$

You should write your answer as: $-\log_2 3$

Reciprocating the input negates the output.

This is known as changing the base. So to get $\log_2 9$ in terms of log base 3:

$log_2 9 = \frac{log_3 9}{log_3 2} = \frac{2}{log_3 2}$

1.22 Exponentials and Logarithms

Chapter 14 · Original slide 29 · Examples

Write as a single logarithm:

a. $\log_3 6 + \log_3 7$

b. $\log_2 15 - \log_2 3$

c. $2\log_5 3 + 3\log_5 2$

d. $\log_{10} 3 - 4\log_{10}\left(\frac{1}{2}\right)$

Write in terms of $\log_a x$, $\log_a y$ and $\log_a z$

a. $\log_a(x^2 y z^3)$

b. $\log_a\left(\frac{x}{y^3}\right)$

c. $\log_a\left(\frac{x\sqrt{y}}{z}\right)$

d. $\log_a\left(\frac{x}{a^4}\right)$

Chapter 14 · Original slide 30 · Anti Laws

These are NOT LAWS OF LOGS, but are mistakes students often make:

$\log_a(b+c) = \log_a b + \log_a c$

There is no method to simplify the log of a sum, only the sum of two logs!

FAIL

$(\log_2 x)^3 = 3\log_2 x$

The power must be on the input (here the x), but here the power is around the entire log.

FAIL

Chapter 14 · Original slide 31 · Solving Equations with Logs

Solve the equation $\log_{10}4 + 2\log_{10}x = 2$

We've used the laws of logs to combine them into one.

The subtle bit: You must check each value in the original equation. We're not allowed to log a negative number.

This is a very common type of exam question.

The strategy is to combine the logs into one and isolate on one side.

Use your favourite method of rearranging. Either do "10 the power of each side" to "undo" the log, or the "insert the 2 between the 10 and $4x^2$" method.

Chapter 14 · Original slide 32 · Test Your Understanding

a

Given that $2\log_2(x+15) - \log_2 x = 6$,

(a) show that $x^2 - 34x + 225 = 0$. (5)

(b) Hence, or otherwise, solve the equation $2\log_2(x+15) - \log_2 x = 6$. (2)

b

Those who feel confident with their laws could always skip straight to this line.

1.22 Exponentials and Logarithms

Chapter 14 · Original slide 33 · Exercise 14E (323-324)

5 a Given that $\log_3(x+1) = 1 + 2\log_3(x-1)$, show that $3x^2 - 7x + 2 = 0$.

b Hence, or otherwise, solve $\log_3(x+1) = 1 + 2\log_3(x-1)$.

Chapter 14 · Original slide 34 · Exercise 14E (323-324)

Challenge:

By Writing $\log_a x = m$ and $\log_a y = n$

Prove that $\log_a x - \log_a y = \log_a\left(\frac{x}{y}\right)$

6 Given that a and b are positive constants, and that $a > b$, solve the simultaneous equations

$a + b = 13 \log_6 a + \log_6 b = 2$

1.22 Exponentials and Logarithms

Chapter 14 · Original slide 35 · Extension Exercises

These are all strictly non-calculator!

1 2

[AEA 2010 Q1b] Solve the equation

$\log_3(x-7) - \frac{1}{2}\log_3 x = 1 - \log_3 2$

[AEA 2008 Q5i] Anna, who is confused about the rules of logarithms, states that

$(\log_3 p)^2 = \log_3(p^2)$

$\log_3(p+q) = \log_3 p + \log_3 q$

However, there is a value for p and a value for q for which both statements are correct. Find their values.

3

[MAT 2007 1I] Given that a and b are positive and

$4(\log_{10} a)^2 + (\log_{10} b)^2 = 1$

what is the greatest possible value of a?

4

[MAT 2002 1F] Observe that $2^3 = 8$, $2^5 = 32$, $3^2 = 9$ and $3^3 = 27$. From these facts, we can deduce that $\log_2 3$ is:

between $1\frac{1}{3}$ and $1\frac{1}{2}$

between $1\frac{1}{2}$ and $1\frac{2}{3}$

between $1\frac{2}{3}$ and 2

none of the above

Chapter 14 · Original slide 36 · Solving equations with exponential terms

Solve $3^x = 20$

This is often said “Taking logs of both sides…”

Solve $5^{4x-1} = 61$

1.22 Exponentials and Logarithms

Chapter 14 · Original slide 37 · Solving equations with exponential terms

Solve $3^x = 2^{x+1}$

Logs in general are great for solving equations when the variable is in the power, because laws of logs allow us to move the power down.

It doesn't matter what base you use to get the final answer as a decimal, provided that it's consistent. You may as well use the calculator's 'log' (no base) key.

Why can we not apply quite the same strategy here?

This then becomes a GCSE-style 'changing the subject' type question. Just isolate x on one side and factorise out.

Chapter 14 · Original slide 38 · Test Your Understanding

1

Solve $3^{2x-1} = 5$, giving your answer to 3dp.

2

Solve $3^{x+1} = 4^{x-1}$, giving your answer to 3dp.

3

Solve $2^x 3^{x+1} = 5$, giving your answer in exact form.

1.22 Exponentials and Logarithms

Chapter 14 · Original slide 39 · Exercise 14F – Y1 Page 325

3 Solve the following equations, giving your answers to 3 significant figures where appropriate.

a $3^{x+1} = 2000$ (2 mar

b $\log_5(x-3) = -1$ (2 mar

4 a Sketch the graph of $y = 4^x$, stating the coordinates of any points where the graph crosses the axes.

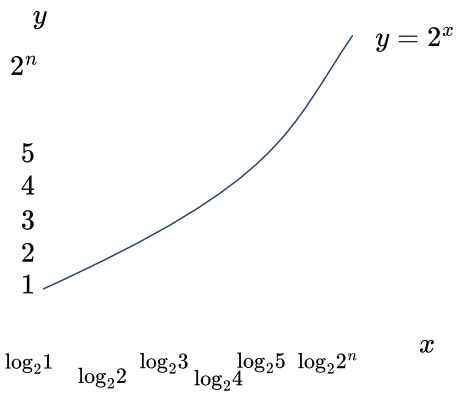
b Solve the equation $4^{2x} - 10(4^x) + 16 = 0$.

1

EXTENSION [MAT 2011 1H] How many positive values x which satisfy the equation:

$x = 8^{\log_2 x} - 9^{\log_3 x} - 4^{\log_2 x} + \log_{0.5} 0.25$

Chapter 14 · Original slide 40 · Very Challenging Extension!



2

[MAT 2013 1J] For a real number x we denote by $[x]$ the largest integer less than or equal to x. Let n be a natural number. The integral

$\int_0^n [2^x] dx$

equals:

(A) $\log_2((2^n - 1)!)$

(B) $n2^n - \log_2((2^n)!)$

(C) $n2^n$

(D) $\log_2((2^n)!)$